

Amjad S Althabteh

[LinkedIn](#) | [Portfolio](#) | [✉ Amjadjamalshadi@gmail.com](mailto:Amjadjamalshadi@gmail.com) | [Github](#) | [734-657-3714](tel:734-657-3714)

Summary

C++ graphics and engine developers focused on real-time rendering and performance-critical systems. Built modular OpenGL engines, GPU-optimized pipelines, and custom allocators. Experienced with Unreal Engine 5 gameplay systems and low-level debugging.

Education

B.S in Computer Science	GPA: 3.65/4.00	Wayne State University	Detroit, MI	2026
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Experience

Graphics Engineer, Co Founder	Immersalab Graphics	Detroit, MI	Sep 2025 - Dec 2025
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- Architected a real-time C++20/OpenGL rendering engine with GPU-driven pipelines, emphasizing frame-time stability, cache efficiency, and performance-critical system design.
- Made an internal UI and JSON-based asset pipelines for flexible content creation.
- Implemented data-driven rendering optimizations for performance balancing.
- Added GPU profiling and frame-timing tools for debugging and optimization.

Unreal C++ Dev	Indie Game (NOA)	Ann Arbor, MI	Apr 2025 - Aug 2025
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- Developed gameplay systems in C++20 and visual scripting, integrating animation and narrative logic.
- Created and rigged 3D environments and characters.
- Produced a 90+ second real-time cinematic using MetaHuman, Control Rig, and Sequencer.
- Designed animation control systems with C++ and Animation Blueprints for smooth blending. [no_one_answers_slicee](#)

Software Development Intern	Rare Munchies	Ann Arbor, MI	Sep 2023 - Apr 2024
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- Integrated UI components using **TypeScript**, improving page transitions and interaction flow.
- Identified and resolved **cross-browser compatibility issues**, reducing debugging time and unblocking delivery
- Optimized client-side rendering logic and state management patterns to reduce unexpected UI re-renders.

Research Experience

Undergraduate Researcher – Orbital Collision Modeling	Wayne State University — Detroit, MI Feb 2025 – Present
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- Modeled satellite orbital trajectories using numerical integration methods and simulated collision probabilities through N-body gravitational dynamics and spatial partitioning techniques.

Projects

3D Space Simulation Engine – C++/OpenGL	space-simulations
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- Engineered custom 3D space simulation engine in C++17 using OpenGL 4.x, GLAD, and SFML.
- Developed modular engine architecture (rendering, physics, math, entities, and universe systems).
- Implemented custom math library (Vec3, Mat4) and camera system with projection/view matrices.

HFT OrderBook Engine C++ 20

- Architected a modular price-time FIFO matching engine with bid/ask depth tracking and partial fill execution logic.
- Designed low-latency infrastructure using cache-aligned data structures and modern C++20 features to minimize branch misprediction.

Glint 3D – C++ Automated 3D Rendering Engine

[glint-rendering](#)

- Optimized a C++17/OpenGL rendering engine with optimized GPU workload and lighting.
- Cross-platform architecture with runtime scene analysis for stable performance.
- Implemented custom C++20 memory pool allocator, reducing heap allocations and improving runtime stability under load.

C++ Runtime Stack Trace & Diagnostics Engine (Traceforge)

- Built a C++23 debugging system for automated stack-trace parsing.
- Automated regression test generation with intelligent fix suggestions.
- Added memory tracking utilities to detect leaks and invalid allocations.

Skill

- **Programming Languages:** C++ (C++11–C++23), C, C#, Python, TypeScript, Java
- **Graphics & Engines:** OpenGL • Unreal Engine 4 - 5.7 • Unity • Godot • NX Design
- **Technical Specialties :** Gameplay Systems, VR/AR Development, Simulation, Tools & Plugins, Technical Art